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Sulfuric acid-mediated PET-derived carbon quantum dots for smartphone-assisted Fe^{3+} screening in biological samples

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Recycling reduces environmental pollution, conserves natural resources, and minimizes waste, while monitoring ferric ions is important for evaluating iron-related biological and environmental conditions. The present work addresses two important challenges: recycling plastic waste and developing a simple sensing platform for Fe^{3+} detection. In this study, post-consumer PET bottles were converted into highly fluorescent carbon quantum dots (PET-CQDs) with an average diameter of 5 ± 2 nm for the detection of ferric ions (Fe^{3+}). The PET-CQDs exhibited Fe^{3+} detection through an ON–OFF fluorescence sensing platform. A linear range of 10–100 μM was achieved, with a correlation coefficient of $R^2 \geq 0.9886$. The detection and quantification limits were 3.66 μM and 11.1 μM, respectively. Additionally, a smartphone-based visual detection method achieved a linear range of 10–70 μM for Fe^{3+}, with a correlation coefficient of $R^2 \geq 0.9877$. Using smartphone-assisted detection, the detection and quantification limits were recorded as 4.4 μM and 13.4 μM, respectively. The ON–OFF sensing platform was further examined in a deproteinized human serum as a proof-of-concept matrix study, showing that the PET-CQDs retained Fe^{3+}-responsive fluorescence behavior under the tested conditions. Furthermore, the greenness of the proposed method was assessed using various green metrics, including ComplexMoGAPI, AGREEprep, and BAGI tools.

1 Introduction

Carbon dots (CDs), also commonly referred to as carbon quantum dots (CQDs), have gained significant attention for their notable optical properties, adaptable surface chemistry, ease of synthesis, and biocompatibility,[1,2] enabling applications in biological imaging,[3–5] LEDs,[6] photocatalysts,[7] drug delivery,[8] and biochemical sensors.[9–13] The adjustable photoluminescence of CQDs, influenced by factors such as size, surface states, and doping with heteroatoms, alongside their excitation-dependent emission across the full color spectrum, positions them as excellent candidates for optical sensing applications.[14,15] A range of synthesis techniques is employed to produce CQDs, these include chemical oxidation, pyrolysis, microwave-assisted, electrochemical, and hydro/solvothermal methods.[16,17] Recent advances have shown that CQDs can be produced on a large scale in some systems, highlighting the broader scalability potential of this class of nanomaterials.[18] Carbon sources used for synthesizing CQDs are diverse and encompass coal, graphite oxide, fruit waste, various plant materials, and discarded polymers.[19–21] Notably, the production of biomass-derived CQDs presents a sustainable option leveraging renewable resources,[22] while transforming low-value waste into these nanomaterials is increasingly acknowledged as an effective strategy for sustainable material processing and resource recovery.[23–25]

Iron is an essential element involved in oxygen transport, enzymatic activity, and several biological processes.[26] However, abnormal iron levels can be associated with health disorders, including anemia, organ dysfunction, oxidative stress, and neurodegenerative diseases.[27] Therefore, monitoring ferric ions (Fe^{3+}) is important in biological and analytical contexts. Traditional methods for Fe^{3+} detection are accurate, but they often rely on complex instrumentation, trained personnel, and laboratory-based procedures.[28–30] Fluorescence-based sensing has attracted considerable attention due to its exceptional sensitivity, simplicity, and capability for real-time detection of target analytes.[31–36] Matrix interferences and the need for practical screening tools continue to motivate the development of simple, low-cost, and portable sensing systems. Fluorescent CQD-based sensors have emerged as promising platforms for Fe^{3+} detection because of their sensitivity, facile signal readout, and potential compatibility with portable analytical formats.[37]

Because many conventional detection methods are slow, complex, and instrument-dependent, the need for fast and real-time analytical approaches remains important.[38] Mobile-enabled sensing platforms are promising alternatives because

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of their portability, user-friendly interfaces, and feasibility as sample-to-answer analytical tools.[32,34,39] Modern smartphones combine high-resolution cameras with built-in computational processing, enabling quantitative analysis without specialized optical instruments.[40] This capability has supported the development of both colorimetric and fluorescence-based detection systems, with fluorescence offering high sensitivity and visual detectability.[41–43] Smartphone-based detection has been demonstrated across diverse analytical applications, offering a strong opportunity for field and point-of-need monitoring.[44] Among these strategies, image-based analysis is particularly appealing because of its low cost, flexibility, and ease of use, especially when colorimetric or fluorescence signals are converted into RGB values for real-time measurement.[45–47]

Over the past 50 years, plastics have replaced materials such as paper, wood, and metal due to their high strength, durability, flexibility, and resistance to environmental stress.[48] Nevertheless, these same properties make plastics highly resistant to biodegradation, resulting in severe and growing global ecological risks.[49–51] Polyethylene terephthalate (PET) has exhibited the fastest growth in the plastics industry, and is predominantly driven by the PET bottle industry.[52] Consequently, a great deal of literature has been dedicated to recycling this plastic waste to develop valuable products.[53] According to the United Nations Environment Programme, 43 percent of plastic waste ends up in landfills, while 22 percent causes direct harm to public health and ecosystems.[54] Converting excess PET into carbonaceous materials has emerged as a promising strategy and an economic opportunity.[55–57]

Recent studies and reviews have demonstrated that waste PET and other discarded materials can be successfully converted into functional carbon quantum dots, highlighting diverse synthesis routes, tunable surface chemistries, and promising analytical applications.[58–65] In particular, PET-derived carbon dots have already been reported for Fe^{3+} fluorescence sensing, including the recent work of Promcharoen *et al.*[66] and the PET-derived carbon dots reported by Wu *et al.*[67] Despite these advances, further work is still needed to combine PET-derived CQDs with simple acid-mediated preparation, portable smartphone-assisted readout, serum-matrix evaluation under controlled conditions, and greenness assessment within a single sensing workflow.

In this context, we developed a one-step solvothermal method for converting waste PET bottles into carbon quantum dots (PET-CQDs) using sulfuric acid as the reaction medium. Building on our previous H_3PO_4-mediated PET-derived carbon dot study,[65] the present work uses H_2SO_4 as a different acidic and dehydrating reaction medium to examine its effect on PET degradation, carbonization, surface functional groups, and fluorescence behavior. The present study is therefore framed as an extension of PET-derived carbon-dot sensing research. Its contribution lies in linking PET valorization, acid-mediated CQD preparation, Fe^{3+}-responsive fluorescence screening, smartphone-assisted image analysis, and green-metric evaluation. The obtained PET-CQDs exhibited concentration-dependent fluorescence quenching in the presence of Fe^{3+} ions, supporting their use as fluorescent probes for proof-of-concept Fe^{3+} screening in a deproteinized serum matrix. This approach emphasizes sustainability through waste reuse, simple synthesis, portable analysis, and environmentally oriented method assessment, resulting in a low-cost and practical fluorescent probe for Fe^{3+}-responsive screening. Scheme 1 summarizes the synthesis route and the on-off sensing mechanism of the PET-CQDs.

2 Experimental section

2.1 Reagents and materials

All chemicals were procured from Sigma-Aldrich with analytical grade purity. These included inorganic acids (hydrochloric acid 36–38%, sulfuric acid 99.9%, nitric acid 68–70%), bases (NaOH), metal salts (ammonium iron(II) sulfate, silver nitrate, aluminum nitrate, cadmium nitrate, sodium acetate, cobalt chloride, chromium chloride, copper nitrate, iron(III) nitrate, mercury(II) nitrate, mercury(I) chloride, potassium iodide, potassium bromide, magnesium chloride, manganese chloride, sodium carbonate, sodium sulfide, sodium sulfate, sodium chloride, nickel sulfate, lead nitrate, and zinc nitrate), and organic compounds (alanine, albumin, ascorbic acid, creatinine, D-glucose, fructose, and glycine). Ultrapure water (18.2 MΩ cm) was exclusively used for all experimental procedures to minimize interference from ionic impurities and ensure reproducibility.

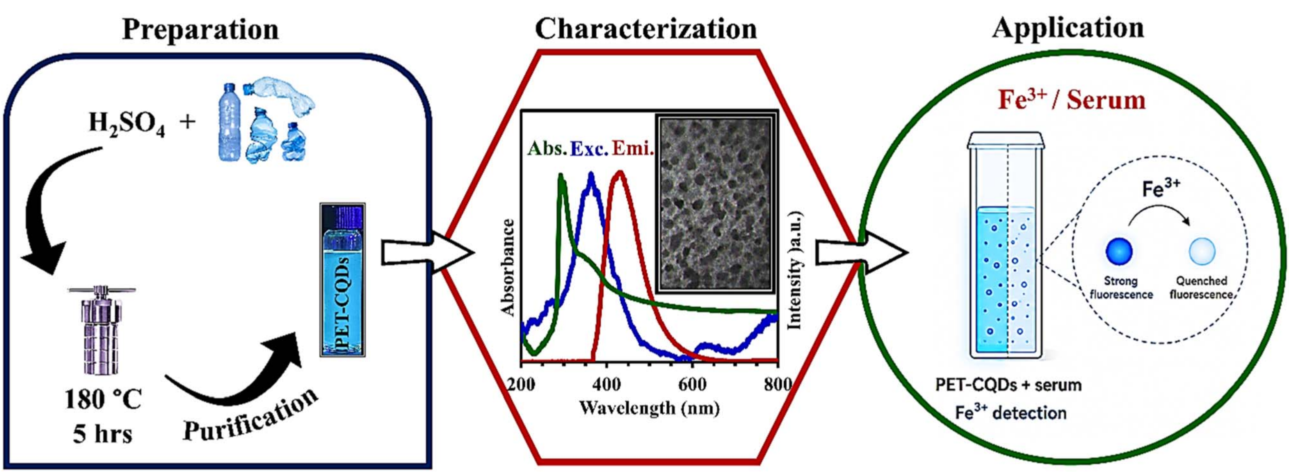
2.2 Characterization

Comprehensive characterization of the synthesized materials was performed using multiple complementary analytical techniques. Ultraviolet-visible (UV-Vis) absorption spectra from 200 to 800 nm were obtained using a Cary 60 Spectrophotometer (Agilent Technologies, USA). Fluorescence emission spectra were obtained using a Cary Eclipse Fluorescence Spectrophotometer (Agilent Technologies, USA), with excitation wavelengths optimized for the carbon quantum dots. Fourier-transform infrared (FTIR) spectroscopy with a Spectrum Two instrument (PerkinElmer) was performed to detect surface functional groups and chemical bonds. The size distribution and morphology of particles were analyzed through transmission electron microscopy (TEM) with an EM10C at 100 kV (ZEISS, Germany). Crystallinity and phase composition were investigated using X-ray diffraction (XRD) with Cu Kα radiation ($\lambda = 1.54$ Å) in an X'Pert Pro system (Panalytical).

2.3 Preparation of PET-CQDs

CQDs were synthesized from polyethylene terephthalate (PET) waste *via* solvothermal decomposition, a sustainable approach that valorizes plastic waste. This method builds on established protocols with modifications to optimize fluorescence properties.[65] To avoid corrosion and contamination, the solvothermal synthesis was performed in a stainless-steel autoclave lined with a Teflon-lined interior.

In a typical synthesis procedure, PET waste bottles are mechanically shredded into fragments of about 2 × 2 mm. 0.2 g of these fragments were placed in 5 mL of concentrated sulfuric



Scheme 1 Schematic illustration of the procedural steps followed in the research methodology.

acid (99.9%), which would serve as both the solvent and the acid catalyst for polymer degradation in the autoclave. Replacing phosphoric acid with sulfuric acid was implemented to examine the effect of a stronger acidic and dehydrating medium on PET degradation, carbonization, and the fluorescence behavior of the resulting PET-CQDs. This sealed autoclave was then heated to 180 °C and held at this temperature for 5 hours, allowing for complete polymer decomposition without over-degradation. After this thermal treatment, the autoclave was cooled naturally to room temperature.

The resulting dark mixture was subjected to a series of purification steps. First, the mixture was passed through a Whatman filter paper to remove any insoluble black residue (unreacted polymer and char). The filtrate was centrifuged at 12 000 rpm for 20 minutes to remove suspended particulates. After that, the final purification was performed using a membrane with a molecular weight cutoff of 1000 Da, which separated carbon quantum dots from smaller molecular byproducts and residual acid through dialysis.

After purification, the PET-CQD solution was adjusted to the acidic working range of pH 2–4 and used as the prepared stock solution for the subsequent fluorescence, stability, Fe^{3+} sensing, serum-matrix, and smartphone-assisted experiments. This pH range was selected based on the pH-stability study, which showed strong and relatively stable PET-CQD fluorescence under acidic conditions. Optimization of synthesis parameters was critical for obtaining high-quality carbon dots. The synthesis conditions used in this study were selected empirically to obtain a homogeneous and strongly fluorescent PET-CQD product suitable for sensing applications. Reaction temperatures in the range of 150–180 °C and different reaction durations were empirically examined to identify conditions that produced a homogeneous and fluorescent PET-CQD solution. A reaction time of 4 hours resulted in incomplete polymer degradation, yielding a solution with reduced fluorescence intensity. Conversely, extending the reaction time to 6 hours led to excessive degradation, resulting in char formation and a dull-colored solution with significantly diminished fluorescence intensity. The optimal conditions of 180 °C for 5 hours provided a balance between complete polymer decomposition and preservation of fluorescence properties, yielding a bright, homogeneous solution suitable for subsequent analytical applications.

2.4 Fluorescence analysis

The fluorescence-quenching properties of the synthesized PET-CQDs were systematically evaluated against a panel of metal cations to identify the ions that produced the strongest fluorescence response and to assess potential interference effects. In the initial screening experiment, 1 mL aliquots of cation solutions (Ag^+, Ca^{2+}, Hg^+, Cd^{2+}, Cr^{3+}, Cu^{2+}, Co^{2+}, Fe^{2+}, Mg^{2+}, Mn^{2+}, Pb^{2+}, Zn^{2+}, Ni^{2+}, Al^{3+}, Fe^{3+}, Hg^{2+}), each at a concentration of 100 μM, were individually mixed with 1 mL of diluted PET-CQDs solution. The resulting mixtures were homogenized and allowed to equilibrate for 20 minutes at room temperature to ensure complete interaction between the metal ions and the CQDs. Then, fluorescence emission intensities were recorded at the characteristic emission wavelength using the Cary Eclipse Fluorescence Spectrophotometer to quantify the quenching efficiency of each cation. For a detailed dose–response analysis, ferric ion (Fe^{3+}) detection was performed at concentrations ranging from 10 to 100 μM (10, 20, 30, 40, 50, 60, 70, 80, 90, 100 μM). Each concentration was prepared by serial dilution from a stock solution and mixed with 1 mL of the diluted PET-CQDs solution. Fluorescence intensities were recorded to establish a calibration curve and determine the linear dynamic range and limit of detection. All Fe^{3+} sensing measurements were performed using the pH-adjusted PET-CQD solution described in Section 2.3.

2.5 Stability of PET-CQDs

The photochemical and environmental stability of the synthesized PET-CQDs was comprehensively evaluated under multiple stress conditions to assess their suitability for practical applications.

2.5.1 Ionic strength stability. A PET-CQDs solution at a constant concentration was added to sodium chloride solutions of varying concentrations to simulate physiological ionic-strength conditions. Fluorescence intensities were measured across the concentration range to assess the PET-CQDs' resistance to ionic quenching and aggregation in saline.

2.5.2 Photostability. A PET-CQDs sample solution was continuously exposed to ultraviolet light (365 nm) for 12 hours. Fluorescence intensity measurements were recorded at 1 hour intervals to quantify photodegradation kinetics and determine the photostability of the fluorescent probe under prolonged UV irradiation.

2.5.3 Thermal stability. The temperature-dependent fluorescence properties of the PET-CQDs solution were evaluated using a temperature-controlled cuvette holder. The solution was subjected to varying temperatures using an ice-water bath (0 °C) and heating to elevated temperatures (up to 80 °C), with fluorescence intensities recorded at regular temperature intervals to assess thermal sensitivity and reversibility of fluorescence changes.

2.5.4 pH stability. The pH-dependent fluorescence behavior of the PET-CQDs solution was systematically investigated by preparing solutions at different pH values (ranging from pH 1 to pH 14). The pH was adjusted using dilute hydrochloric acid and sodium hydroxide solutions, and fluorescence intensities were recorded at each pH value to characterize the ionization behavior of surface functional groups and their influence on photoluminescence.

2.6 Smartphone-assisted analysis

Smartphone-assisted analysis offers a portable, cost-effective, and user-friendly platform for rapid data acquisition and processing, enabling on-site and real-time analytical measurements.[68–74] A portable, cost-effective analytical platform was developed utilizing smartphone imaging technology for on-site detection of metal ions. The smartphone camera used was an iPhone Pro Max equipped with a 48-megapixel primary sensor (f/1.78 aperture), supplemented by 12-megapixel ultra-wide (f/2.2) and 12-megapixel telephoto (f/1.78) lenses, enabling high-resolution imaging of fluorescence responses.

2.6.1 Sample preparation and measurement. Analyte solutions (Fe^{3+}) were prepared at concentrations from 10 to 80 μM by serial dilution from standardized stock solutions. Each analyte concentration was mixed with a fixed concentration of PET-CQDs solution in a standard 1 cm path-length quartz cuvette. Upon interacting with the cation, the CQDs exhibited a distinct, quantifiable change in fluorescence intensity. All smartphone-assisted measurements were conducted under controlled environmental conditions: room temperature (25 ± 3 °C) and using the pH-adjusted PET-CQD solution described in Section 2.3.

2.6.2 Image acquisition. The smartphone camera was positioned at a fixed distance from the cuvette (typically 10 cm) using a custom-designed holder to ensure consistent geometry and minimize parallax errors. Images were captured under standardized lighting conditions using a handheld 360 nm UV lamp positioned at a fixed angle to provide uniform illumination and eliminate shadows or reflections that could compromise image quality.

2.6.3 Image analysis. Quantitative analysis of the fluorescence response was performed using ImageJ software (National Institutes of Health, USA). The blue (B) channel was selected for study as it demonstrated the highest sensitivity to fluorescence variations and provided the best signal-to-noise ratio. A fixed region of interest (ROI) was defined at the center of each cuvette image, with dimensions of 50×50 pixels, to ensure consistent sampling across all measurements and to minimize edge effects and optical artifacts from cuvette walls and reflections. The mean pixel intensity within the ROI was measured for each image. The net fluorescence intensity was calculated by subtracting the background signal obtained from a blank sample (PET-CQDs solution without Fe^{3+}) from the measured intensity of each Fe^{3+}-containing sample. These corrected intensity values were subsequently correlated with the corresponding Fe^{3+} concentrations by linear regression to generate a calibration graph, enabling quantitative determination of unknown samples. This smartphone-based approach provides a simple proof-of-concept portable readout for Fe^{3+} screening under fixed imaging conditions. Broader validation using different smartphone models, camera settings, lighting conditions, imaging distances, and intra-day/inter-day precision studies would be required before the method can be considered a standardized quantitative platform.

2.7 Serum sample collection and pretreatment

A serum sample were obtained as anonymized residual specimen from routine clinical laboratory testing at a Shar Hospital in Sulaimani City, Kurdistan Region, Iraq. No personal identifiers or clinical information were collected or used in this study, and the sample was handled anonymously for method-evaluation purposes following local institutional procedures for residual clinical specimens. No formal ethics approval number or approval letter is available for this work. The serum sample underwent standardized pretreatment procedures as established in the literature.[27] Equal volumes of serum and ethanol were combined and heated at 60–70 °C for 15 minutes to precipitate proteins and denature the interfering biomolecules. The heated mixture was cooled to room temperature and centrifuged at 10 000 rpm for 20 minutes to separate the protein precipitate from the supernatant. The clear supernatant was carefully collected and retained for subsequent analysis. To ensure complete oxidation of all iron species to the ferric form (Fe^{3+}), which is necessary for consistent detection, 1 mL of concentrated nitric acid (6 M HNO_3) was added to the supernatant, and the mixture was allowed to stand for 30 minutes. This oxidation step was used to convert iron species to the ferric form and provide a consistent Fe^{3+} response under the tested conditions.

3 Results and discussion

3.1 Characterization of PET-CQDs

Many analytical techniques were employed to characterize the synthesized PET-CQDs to elucidate their structural, morphological, and compositional properties. These characterization methods provided critical insights into the quality and suitability of the carbon quantum dots for subsequent analytical applications.

XRD analysis was utilized to investigate the crystalline structure and phase composition of the PET-CQDs. The XRD pattern of PET-CQDs displayed a broad diffraction peak centered at approximately 20°, which is indicative of the amorphous nature of the carbon core,[75,76] confirming the successful conversion of the crystalline PET polymer precursor into disordered carbon structures through the solvothermal decomposition process. The absence of sharp, well-defined diffraction peaks further substantiates the amorphous character of the synthesized PET-CQDs, which is consistent with the expected structural properties of CQDs derived from polymer waste.[77]

FTIR spectroscopy was conducted to identify and characterize the surface functional groups and chemical composition of the PET-CQDs. The FTIR spectrum revealed several characteristic absorption bands indicating the presence of oxygen-containing functional groups on the PET-CQDs surface. A broad, intense absorption band centered at approximately 3500 cm^{-1} is attributed to the O–H stretching vibration of hydroxyl groups, indicative of alcohol and carboxylic acid moieties.[78] The strong absorption peak at 1631 cm^{-1} is assigned to the C═O stretching vibration, characteristic of the carbonyl group in carboxylic acids.[79] Additional significant peaks were observed at 1218 cm^{-1} and 1050 cm^{-1}, attributed to C–O stretching vibrations of ether and ester linkages, as well as C–H bending modes.[80,81] These spectroscopic findings indicate the presence of multiple oxygen-containing functional groups on the surface of the PET-CQDs. The abundance of these hydrophilic surface functionalities likely contributes to the water dispersibility of the PET-CQDs and provides potential active sites for interaction with target analytes. Within the scope of the present study, this characterization provides an adequate basis for discussing the surface chemistry relevant to the observed fluorescence behavior and metal-ion sensing performance.

TEM was employed to directly visualize the morphology, size distribution, and structural features of the synthesized PET-CQDs. As shown in Fig. 1C and D, the particle size analysis, based on measurements of multiple individual particles from the TEM images, demonstrated that the PET-CQDs have a size range of approximately 3–20 nm, with a mean diameter of 5 ± 2 nm, as illustrated in the histogram. This size distribution is consistent with the quantum-confinement regime, where particles in this dimension range exhibit quantum-size effects that are responsible for the unique photoluminescent properties of CQDs.[82] TEM images confirm that the solvothermal method enabled the formation of a small-sized quantum domain region.

Overall, the XRD, FTIR, TEM, UV-Vis absorption, and fluorescence results confirm the successful conversion of PET waste into fluorescent PET-CQDs with amorphous carbon structure, nanoscale morphology, and oxygen-containing surface functional groups. These features provide an appropriate basis for discussing the fluorescence behavior and Fe^{3+}-responsive sensing performance observed in this study. Further advanced surface and photophysical analyses, such as XPS, Raman spectroscopy, fluorescence lifetime, quantum yield, and zeta potential measurements, could provide additional insight into the detailed structure–property relationship of PET-CQDs in future work.

3.2 Optical properties of PET-CQDs

The optical properties of the synthesized PET-CQDs were characterized using ultraviolet-visible absorption and fluorescence spectroscopy to elucidate their photoluminescence behavior. Fig. 2A illustrates the UV-visible spectrum of PET-CQDs, which displayed characteristic absorption features indicative of the electronic structure of the CQDs.[83] The spectrum exhibited a prominent absorption peak at approximately 290 nm, which is assigned to the π–π transition of aromatic C═C bonds within the sp^2 carbon framework of the carbon core. Additionally, a secondary absorption band was observed at approximately 350 nm, attributed to the n–π transition of carbonyl (C═O) groups present on the surface of the PET-CQDs.[84,85] These absorption features are consistent with the oxygen-containing functional groups present on the PET-CQDs surface, and the multiple absorption bands reflect the heterogeneous electronic structure of the CQDs, arising from the combination of the sp^2 carbon core and surface functional groups.

The fluorescence properties of the PET-CQDs were systematically investigated through both emission and excitation spectroscopy. The fluorescence emission spectrum showed a peak at 430 nm, characteristic of blue-emitting CQDs. The corresponding excitation spectrum showed a maximum at 360 nm (Fig. 2A), indicating optimal photon absorption at this wavelength for efficient fluorescence generation.[86] The Stokes shift, calculated as the difference between the excitation maximum (360 nm) and the emission maximum (430 nm), was approximately 70 nm, which is typical for CQDs and reflects the energy loss during relaxation from the excited state to the ground state.[87]

A notable optical characteristic of the PET-CQDs was their pronounced excitation-dependent emission behavior, in which fluorescence emission wavelength and intensity varied with excitation wavelength. As the excitation wavelength increases from 310 nm to 390 nm, the emission peak of PET-CQDs shifts to the red, with the intensity initially rising and then gradually decreasing (Fig. 2B). The excitation-dependent photoluminescence observed in the PET-CQDs is a well-documented phenomenon in CQDs and arises from several interconnected mechanisms.[88] One major factor is the heterogeneous size distribution of the synthesized dots (3–20 nm, mean 5 ± 2 nm), which results in selective excitation of particles of different sizes due to quantum size effects. Smaller dots preferentially absorb shorter-wavelength photons and emit in the blue region, whereas larger dots absorb and emit at longer wavelengths.[89] In addition, multiple emissive surface sites, originating from diverse oxygen-containing functional groups such as hydroxyl, carboxyl, and ether groups identified by FTIR, contribute to heterogeneous photoluminescence by providing electronic

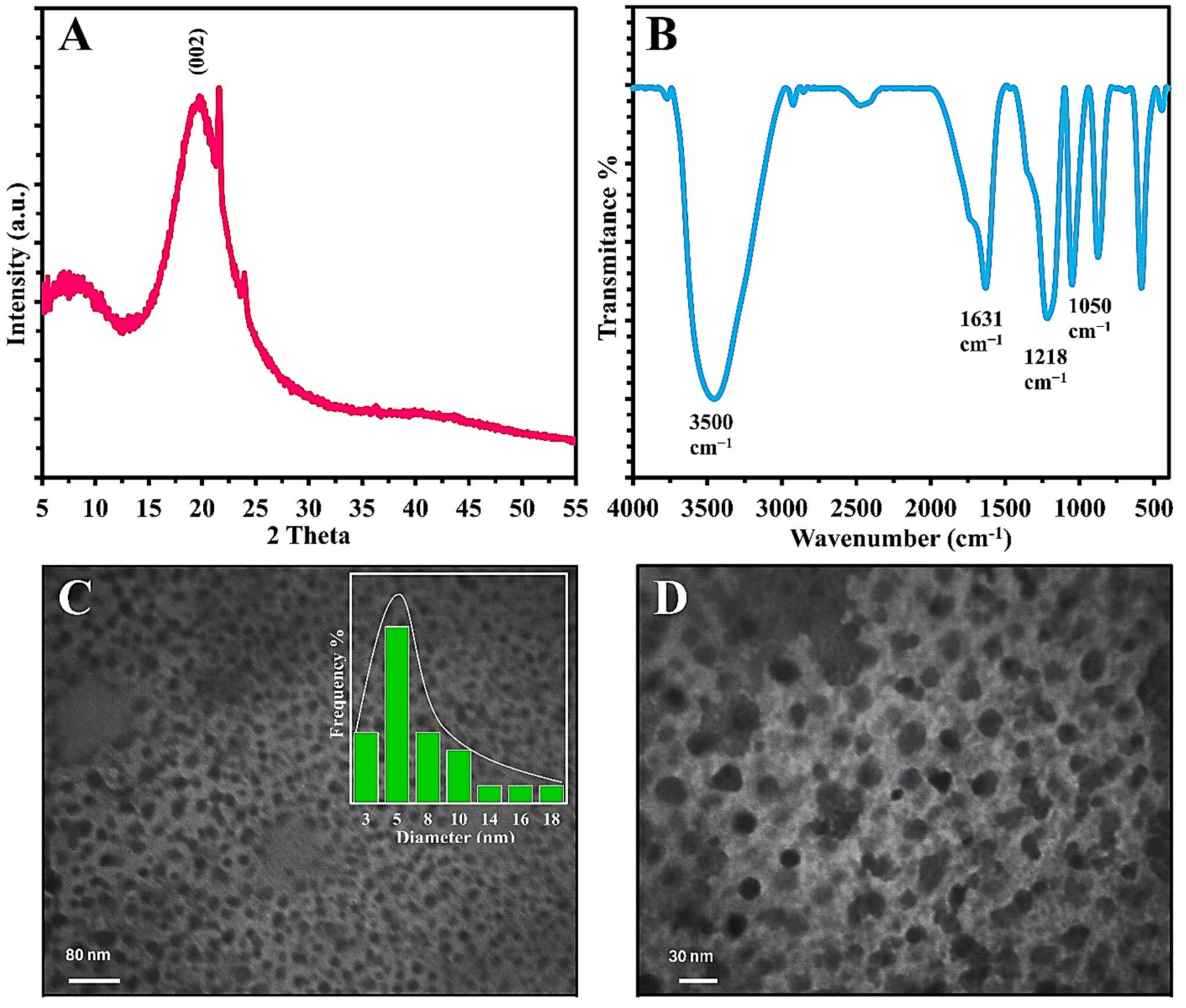


Fig. 1 (A) X-ray diffraction pattern of PET-CQDs. (B) FT-IR spectra of PET-CQDs. (C) and (D) TEM images of PET-CQDs. The particle-size distribution histogram is shown in the inset.

states that can be selectively excited at different wavelengths.[86] The electronic structure of the carbon core, consisting of sp^2 carbon clusters dispersed within an sp^3 matrix, also reinforces excitation dependence because variations in the conjugation and distribution of these sp^2 domains result in wavelength-dependent absorption and emission pathways.[90] Finally, solvent relaxation dynamics, notably slower dielectric relaxation around the PET-CQDs, contribute to inhomogeneous spectral broadening and spectral migration, which enhances the appearance of excitation-dependent emission.[91]

3.3 Stability of PET-CQDs

The photochemical and environmental stability of fluorescent probes is a critical parameter for their practical application in analytical systems. Comprehensive stability assessment of the synthesized PET-CQDs was therefore conducted under multiple stress conditions, including high ionic strength, prolonged photochemical exposure, temperature variations, and pH extremes, to evaluate their robustness and suitability for real-world analytical applications.

The resistance of PET-CQDs to ionic quenching and aggregation in high-salt environments is essential for applications in physiological and environmental samples.[92] As illustrated in Fig. 2C, the fluorescence intensity of PET-CQDs remained largely unaffected when mixed with sodium chloride solutions across a concentration range up to 1.0 M. This ionic strength independence proves that the PET-CQDs fluorescence is not significantly affected by electrostatic interactions or aggregation in the presence of common inorganic ions. Such stability is especially beneficial for applications involving serum analysis, environmental water samples, and other complex matrices with high salt concentrations.

Photochemical stability is essential for fluorescent probes used in continuous or long-term monitoring.[93] The photostability of the PET-CQDs was assessed under continuous high-

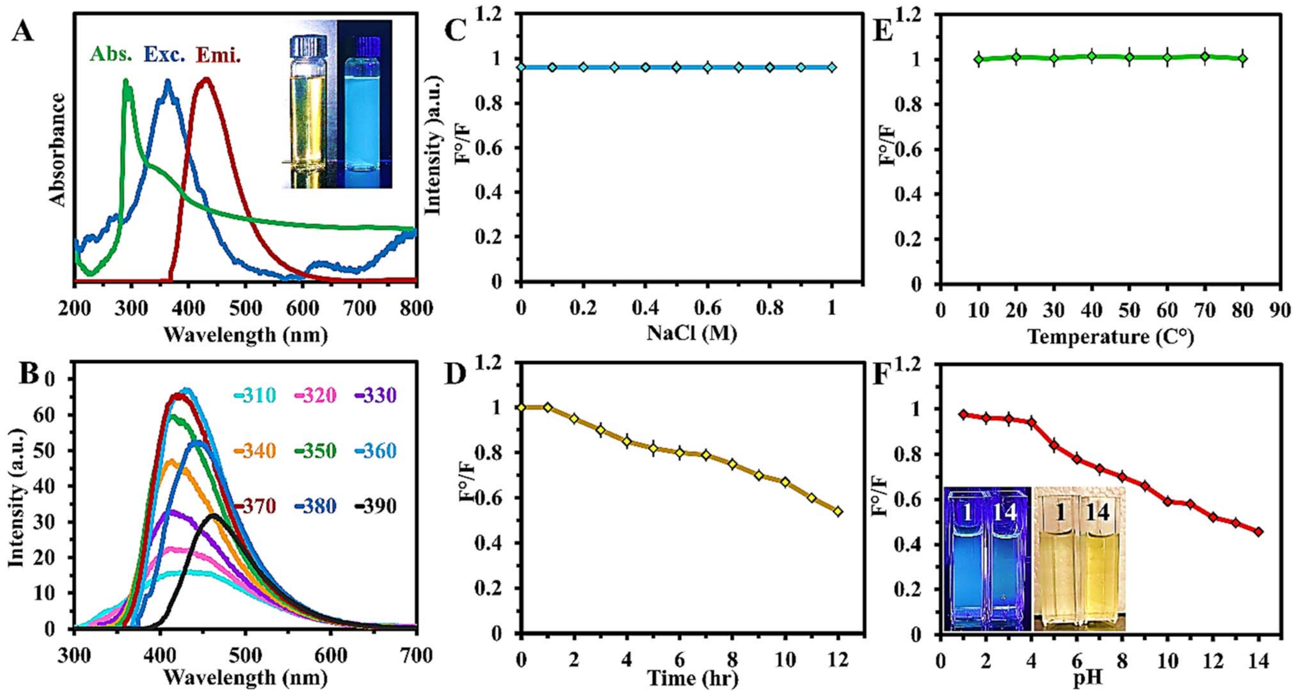


Fig. 2 (A) Absorption spectra, excitation spectra, and fluorescence emission spectra of PET-CQDs; inset: images of PET-CQDs under visible light (left) and UV light (right). (B) Emission spectra of PET-CQDs under varying excitation wavelengths (indicated in the figure). Effects of (C) ionic strength; (D) prolonged exposure to an 8 watt ultraviolet lamp emitting light at a λ of 365 nm on PET-CQDs; (E) temperature; and (F) pH, inset: images of PET-CQDs under visible (right) and UV-light (left) in pH of 1 and 14. All measurements were performed in triplicate. Data in C–F are presented as mean $\pm$ SD ($n = 3$), while representative spectra are shown in A and B.

intensity UV irradiation (365 nm, 8 W) for 12 hours. As shown in Fig. 2D, the fluorescence intensity gradually decreased but retained approximately 50% of its initial value after prolonged irradiation, indicating moderate photostability under the tested harsh illumination conditions. The retention of fluorescence after extended UV exposure suggests that the PET-CQDs have sufficient photochemical stability for repeated analytical measurements within the experimental timeframe. This stability can be attributed to the robust carbon core and surface functional groups, which may help limit photoinduced degradation.

Temperature fluctuations can negatively affect fluorescence-based measurements, especially in field settings where temperature control is limited.[94] The thermal stability of the PET-CQDs was evaluated from 10 °C to 80 °C, spanning both low and elevated temperatures. As shown in Fig. 2E, the fluorescence intensity remained essentially unchanged across this entire range, demonstrating substantial temperature insensitivity. This stability eliminates the need for strict temperature regulation during analysis and supports reliable performance in varied environments. The robustness likely arises from the strong carbon core and stable surface functional groups, which resist structural and ionization changes across the tested temperature range. Overall, this thermal stability enhances the suitability of PET-CQDs for both field-based and laboratory applications.

The fluorescence of PET-CQDs is strongly influenced by pH due to the ionization of surface functional groups.[95] The pH-dependent behavior of PET-CQDs was examined across a wide pH range, and the results are shown in Fig. 2F. The fluorescence intensity was highest at acidic conditions (pH 2–4) and remained stable below pH 4. At neutral and alkaline pH, the intensity decreased, with a notable ~50% drop at pH 14, indicating significant quenching under strongly alkaline conditions. Based on this behavior, the pH range of 2–4 was selected as the working acidic range for Fe^{3+} determinations, since it provided stronger and more stable PET-CQD fluorescence response.

The pH-dependent fluorescence of CQDs is primarily governed by the protonation and deprotonation of surface functional groups, including carbonyl (C=O), carboxyl (–COOH), and hydroxyl (–OH) groups. These ionizable groups interact dynamically with pH variations, creating pH-sensitive fluorescence through surface defect effects and changes in the electronic structure of the carbon dots.[96,97]

CQDs generally consist of a graphitic or amorphous carbon core surrounded by various reactive surface groups, leading to multiple interpretations of their pH-responsive behavior. For PET-CQDs, this pH sensitivity is primarily governed by the protonation/deprotonation of surface functional groups, as illustrated in Fig. 3A. In acidic environments, carboxyl groups are protonated to –COOH, which reduces electrostatic repulsion, suppresses non-radiative decay pathways, and stabilizes dispersion, resulting in strong fluorescence. Under neutral and alkaline conditions, these groups deprotonate to $–COO^-$, generating negative surface charges that increase electrostatic repulsion, promote non-radiative relaxation through electron–phonon coupling, and potentially enable charge-transfer interactions with cations. These effects contribute to fluorescence quenching, with the pronounced intensity loss at pH 14 indicating extensive deprotonation and possible hydroxide-induced surface modifications or partial structural degradation that further diminish photoluminescence.

Overall, the stability evaluation shows that PET-CQDs exhibit strong robustness across key environmental parameters, including ionic strength, light exposure, temperature, and pH. This combination of properties makes them a reliable and versatile fluorescent probe for a wide range of analytical applications. Although their fluorescence is pH dependent, this behavior can be controlled through buffer selection or used advantageously for pH sensing. These stability features support the suitability of PET-CQDs for ferric ion detection applications discussed in the following sections.

3.4 Fluorescence analysis and interference study

The fluorescence response of PET-CQDs toward Fe^{3+} and other potentially interfering species was evaluated to establish their suitability as Fe^{3+}-responsive probes for biological sample analysis. The interference study was performed using a panel of metal ions, common anions, and biomolecules under identical experimental conditions. The results showed that Fe^{3+} produced a pronounced fluorescence quenching response, indicating a strong interaction between Fe^{3+} ions and the PET-CQD surface under the tested conditions.

Quenching responses were measured for a panel of cations, including Ag^{+}, Ca^{2+}, Hg^{+}, Cd^{2+}, Cr^{3+}, Cu^{2+}, Co^{2+}, Fe^{2+}, Mg^{2+}, Mn^{2+}, Pb^{2+}, Zn^{2+}, Ni^{2+}, and Al^{3+}, and Hg^{2+}. Among the tested metal ions, Hg^{2+} also produced a strong quenching response under the tested single-ion conditions, indicating that PET-CQDs can interact with Hg^{2+} as well. However, because the analytical focus of the present study is Fe^{3+} detection in serum

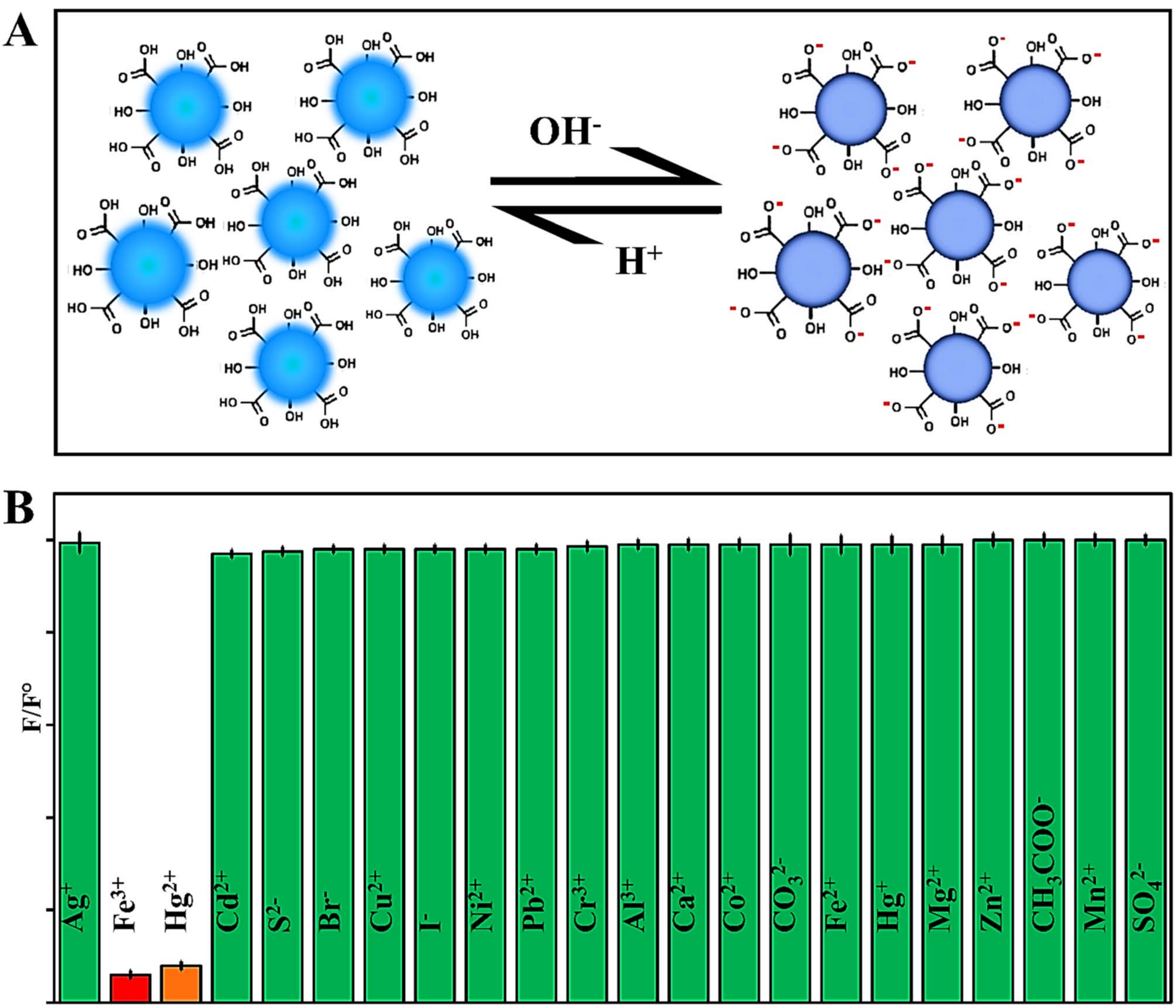


Fig. 3 (A) Scheme shows the deprotonation and protonation of the PET-CQDs. (B) Recorded fluorescence intensities of the PET-CQDs when introduced to a range of ionic species. Data are presented as mean ± SD from three replicate measurements ($n = 3$).

samples, Hg^{2+} is considered a cross-responsive interferent rather than a target analyte in this work. This distinction is important for defining the scope of the proposed sensing platform, particularly in samples where Hg^{2+} may be present.

In contrast, the other tested cations caused only minor changes in fluorescence intensity under the same conditions, indicating that the PET-CQDs have good tolerance toward a broad range of common metal ions. Additionally, a range of common anions (SO_4^{2-}, CO_3^{2-}, CH_3COO^-, I^-, Br^-, S^{2-}) and biomolecules (albumin, alanine, ascorbic acid, creatinine, D-glucose, fructose, and glycine) were also evaluated for potential quenching effects. None of these species produced significant fluorescence suppression under the single-species screening conditions, suggesting reasonable tolerance toward these common matrix components.

Fig. 3B summarizes the selectivity results obtained from single-ion testing at a fixed concentration (PET-CQDs concentration fixed; analyte concentration 100 μM; 20 min equilibration). The pronounced response toward Fe^{3+} supports the use of PET-CQDs as Fe^{3+}-responsive fluorescent probes for controlled biological analysis. At the same time, the observed Hg^{2+} response highlights a limitation of the system and indicates that the proposed method is most appropriate for Fe^{3+} analysis in Hg^{2+}-free or controlled matrices, such as serum samples as examined in this work. For samples in which Fe^{3+} and Hg^{2+} may coexist, additional sample pretreatment, separation, masking, or an independent confirmatory method would be required before accurate Fe^{3+} quantification. Any masking approach should also be experimentally validated to confirm that it suppresses Hg^{2+} interference without affecting the Fe^{3+} response or the fluorescence stability of PET-CQDs.

3.5 Fluorometric and smartphone-assisted Fe^{3+} detection

To evaluate the sensing performance of PET-CQDs toward Fe^{3+}, we investigated the effect of stepwise additions of Fe^{3+} ions on the fluorescence of PET-CQDs under identical experimental conditions. Fig. 4A shows representative fluorescence spectra illustrating the concentration-dependent quenching of PET-CQDs by Fe^{3+}. The spectra displayed in the main figure were selected to provide a clear visualization of the quenching trend, while the calibration analysis was based on the measured concentration series within the reported range.

The fluorescence response showed good linearity over the Fe^{3+} concentration range of 10–100 μM, with $R^2 = 0.9886$. Blank measurements containing PET-CQDs without Fe^{3+} were used to estimate the standard deviation of the blank response, while the slope of the calibration curve was used to calculate the LOD and LOQ according to 3Sd/k and 10Sd/k, respectively. Based on this approach, the LOD and LOQ were calculated as 3.66 μM and 11.1 μM. This sub-10 μM calculated LOD, together with the working linear range, supports the applicability of PET-CQDs for Fe^{3+} screening under the tested analytical conditions.[98,99]

To further evaluate the potential of PET-CQDs for on-site detection, we extended the sensing workflow to include a smartphone-based visual readout. Reaction mixtures containing varying concentrations of Fe^{3+} were prepared as described earlier, placed in cuvettes, and photographed under UV illumination using a smartphone camera to emulate portable detection in resource-limited environments (Fig. 4C). The calibration plot shown in the inset of Fig. 4B was generated from RGB values extracted from the actual smartphone-captured images using the ImageJ protocol described in the Methods section. All images were acquired under fixed experimental conditions, including controlled distance, illumination, and sample positioning, to minimize variability and establish a standardized proof-of-concept readout. Among the extracted RGB components, the blue channel was selected for quantitative analysis because it provided the most suitable and consistent concentration-dependent response under the tested conditions. In addition to the blue component, reductions in the red and green channels were also observed with increasing analyte concentration, reflecting broad-spectrum quenching behavior that aligns with previously documented carbon-dot sensing mechanisms.[100,101]

Using this smartphone-assisted approach, a linear range of 10–70 μM was observed for Fe^{3+} ions ($R^2 = 0.9877$). A detection limit of 4.4 μM and a quantification limit of 13.4 μM were recorded. The smartphone-assisted calibration followed the same concentration-dependent quenching trend observed by conventional fluorometry, supporting its use as a simple visual screening approach. However, the smartphone data were obtained using one phone model under fixed imaging conditions. Therefore, additional testing with different phone models, lighting conditions, camera settings, distances, and intra-day/inter-day measurements would be required for full method standardization.

3.6 Fe^{3+} quenching mechanism

Emission-quenching Stern–Volmer analysis is a pivotal tool for elucidating the interaction between PET-CQDs and quencher ions, revealing the mechanism and efficiency of fluorescent quenching. In a typical Stern–Volmer experiment, emission intensity is measured with a fluorometer, and solutions containing a constant fluorophore concentration and varying quencher concentrations are prepared under controlled conditions to minimize external quenching effects.[102]

As the quencher concentration increases, the excited states of carbon dots can undergo processes such as energy transfer or photoinduced electron transfer, which compete with radiative decay and reduce photon emission. This decrease in fluorescence intensity is described by the Stern–Volmer equation:

$$\frac{F_0}{F_1} = 1 + K_{SV}\ [Q] \tag{1}$$

Here, F_0 and F_1 represent the fluorescence intensities in the absence and presence of the quencher, respectively, while K_{SV} is the Stern–Volmer quenching constant.

The fluorescence quenching behavior of PET-CQDs by Fe^{3+} was studied by plotting the relative change in fluorescence intensity (F_0/F_1) as a function of the quencher's concentration ([Q]). Solutions with varying concentrations of Fe^{3+} and a constant PET-CQD concentration were prepared under

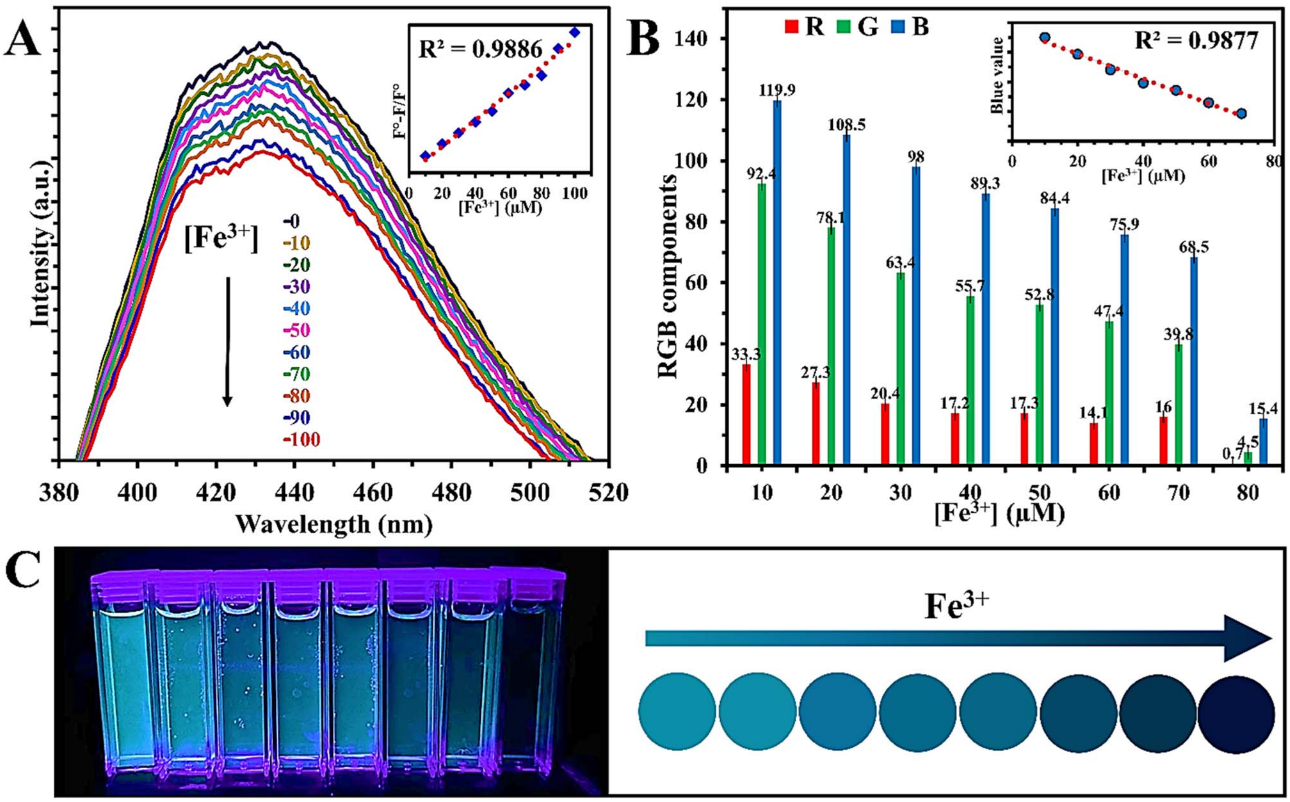


Fig. 4 (A) Fluorescence emission spectra of PET-CQDs following mixing with varying concentrations of Fe^{3+} (Inset: The correlation between Fe^{3+} concentrations ranging from 10 to 100 μM and the fluorescence quenching efficiencies of PET-CQDs.). (B) Concentration-dependent changes in RGB channel intensities extracted from the photographed samples using ImageJ (Inset shows the linear calibration obtained from the blue-channel response under fixed imaging conditions). (C) Smartphone-captured photos of PET-CQDs with various concentrations of Fe^{3+} under UV illumination. Quantitative data are presented as mean ± SD from three replicate measurements (n = 3).

identical conditions, and their emission intensities were measured individually to construct the Stern–Volmer plot, as shown in Fig. 5A.

Linear Stern–Volmer behavior is typically associated with either dynamic (collisional) quenching, static quenching *via* ground-state complex formation, or a combination of both, whereas upward-curving or composite plots indicate concurrent static and dynamic processes or more complex interaction pathways.[103] In the case of PET-CQDs, this behavior implies that, in addition to forming a ground-state complex, some PET-CQDs are quenched due to collisional interactions, without being involved in ground-state complex formation.[104] The surface of PET-CQDs is rich in functional groups, which can introduce numerous defects that act as traps for excitation energy. These surface groups can interact with Fe^{3+} ions, leading to electron transfer and fluorescence quenching.

Possible ground-state association is supported by UV-Vis spectral changes. As shown in Fig. 5B, the addition of the Fe^{3+} ions to PET-CQDs induces discernible modifications in the absorption spectra, consistent with the formation of PET-CQD–ion ground-state complexes. The comparative spectra of PET-CQDs, PET-CQDs/Fe^{3+} (PET-CQDs in the presence of Fe^{3+}), and Fe^{3+} ions show spectral changes consistent with an interaction between Fe^{3+} and PET-CQDs. These changes support the possibility of a static quenching contribution, although they do not provide definitive proof without fluorescence lifetime measurements. Fig. 5C illustrates a schematic representation of a possible quenching pathway, highlighting the formation of the non-fluorescent ground-state complex in addition to the collisional quenching mechanism.

The quenching behavior observed upon addition of Fe^{3+} ions may involve more than one pathway, including surface interaction, possible ground-state association, electron transfer, and collisional quenching. Within the scope of the present study, the mechanism proposed should be considered a plausible interpretation based on steady-state fluorescence and UV-Vis data. Fluorescence lifetime measurements would be required in future work to distinguish the relative contributions of static and dynamic quenching more confidently.

3.7 Fe^{3+} response in serum samples

The demonstrated sensitivity of PET-CQDs for Fe^{3+} detection prompted evaluation in a real biological matrix to assess practicality, robustness, and potential matrix effects. The real-sample study focused on Fe^{3+} detection in human serum. These investigations were conducted under conditions that reflect typical sample preparation requirements for biological analysis, including serum deproteinization to reduce matrix

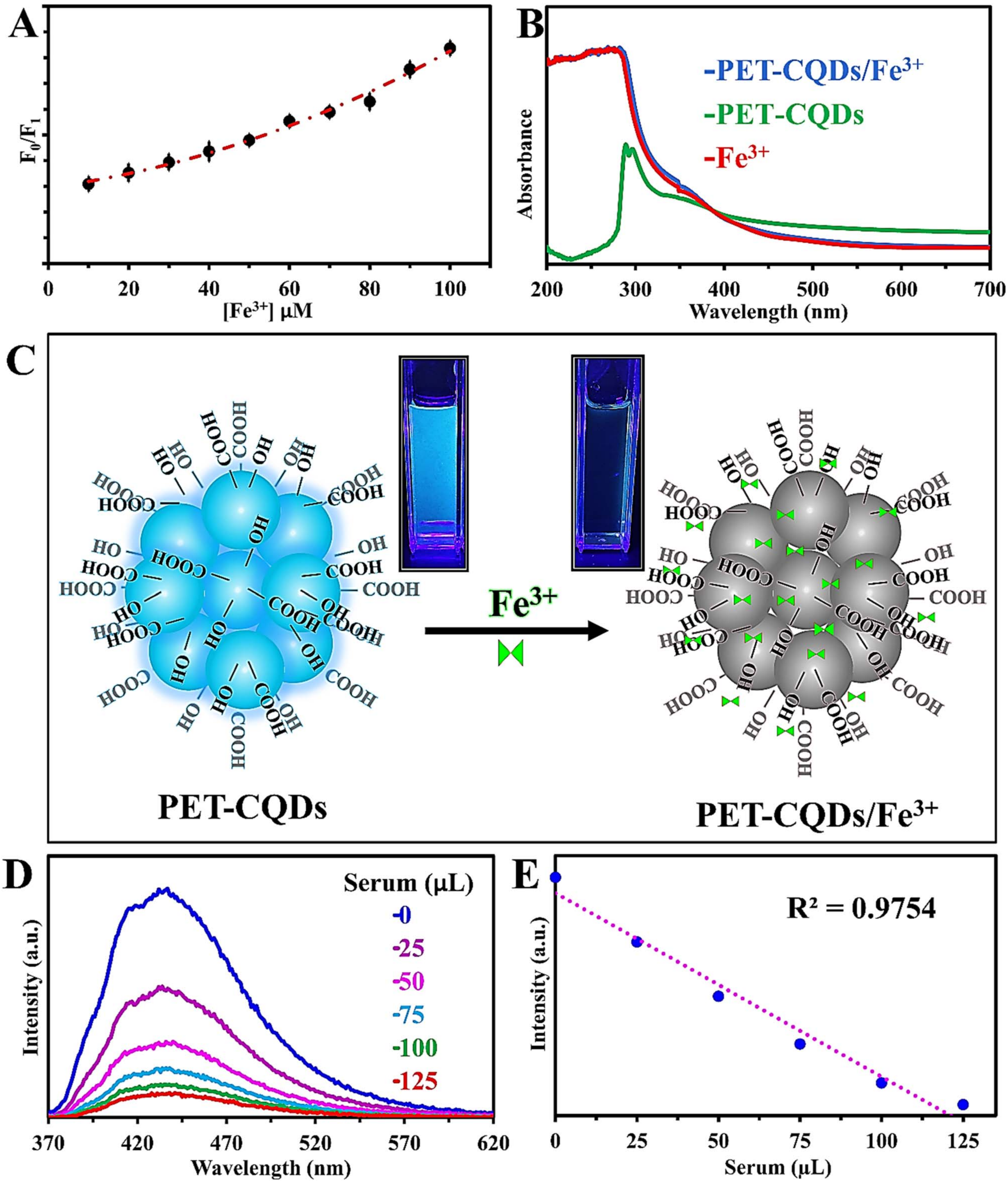


Fig. 5 (A) Stern–Volmer (S–V) plots of PET-CQDs at various concentrations of Fe^{3+}. (B) UV-Visible absorption spectra of PET-CQDs, PET-CQDs/Fe^{3+}, and Fe^{3+}. (C) Scheme showing fluorescence quenching mechanism of PET-CQDs by Fe^{3+} ions. (D) PET-CQDs' fluorescence emission spectra after adding different volumes of the same serum sample. (E) Linear correlation between intensity and serum volume. Quantitative data are presented as mean ± SD from three replicate measurements ($n = 3$), while spectra are shown as representative measurements.

complexity and minimize protein-related interference. Because the present work focuses on Fe^{3+} detection, the method is intended for controlled serum analysis, and samples containing strong cross-responsive metal ions would require additional pretreatment, separation, masking, or confirmatory analysis before accurate Fe^{3+} quantification.

The serum experiment was performed to examine the response of PET-CQDs in a treated biological matrix. As shown in Fig. 5D, incremental additions of diluted deproteinized serum to PET-CQDs caused a gradual decrease in fluorescence intensity, producing a clear correlation between serum volume and quenching efficiency (Fig. 5E). This result indicates that the PET-CQDs retained their fluorescence response in the serum matrix after deproteinization and supports their potential use as a simple Fe^{3+}-responsive screening probe under the tested conditions.

3.8 Greenness evaluation

The core principle of green chemistry is to develop processes that support sustainable development goals. In recent years, increasing attention has been given to environmentally responsible analytical methods in quality control laboratories, technology development, and industrial applications. A crucial aspect of this effort is selecting suitable tools to assess the greenness and practical sustainability of analytical procedures. Several assessment tools have been introduced for this purpose, although each has its own scope and limitations.[105–108] In this study, the greenness evaluation was performed to assess the environmental and practical sustainability of the PET-CQD preparation and sensing workflow. ComplexMoGAPI, AGREEprep, and BAGI were used as complementary tools to evaluate different aspects of the proposed method, including sample preparation, reagent use, operational simplicity, and practical applicability.

ComplexGAPI, an extension of the Green Analytical Procedure Index (GAPI), evaluates the environmental impact of an analytical procedure across all stages, from sample preparation to final analysis, including transportation, preservation, and storage. It features a color-coded pictogram, where additional pre-analytical criteria are represented as hexagonal sections that turn green when sustainability requirements are met.[109] Building upon this, ComplexMoGAPI enhances the evaluation by incorporating a cumulative scoring system that provides a more comprehensive measure of a method's overall environmental sustainability. In addition to the red, yellow, and green indicators used in ComplexGAPI, this new tool offers a more detailed assessment, refining the evaluation process and making it more informative.[110] The ComplexMoGAPI assessment for the PET-CQD method is shown in Fig. 6A and reflects a relatively favorable sustainability profile across key methodological steps.

AGREEPrep, another open-source software, employs a circular pictogram resembling a clock, with 10 impact categories. Each section is rated from 0 to 1, transitioning from deep red (poor sustainability) to deep green (high sustainability), with the final score displayed at the center.[111] The AGREEPrep output for the current method, shown in Fig. 6B, corroborates the favorable sustainability profile indicated by ComplexGAPI and ComplexMoGAPI, highlighting efficient sample-handling steps with reduced procedural burden.

To complement these environmental evaluations, the Blue Applicability Grade Index (BAGI) was used to assess the method's practical suitability for routine analysis. BAGI assigns scores from 25 to 100, with higher scores indicating greater reliability and suitability for routine analysis.[112] As depicted in Fig. 6C, the PET-CQD method received a high BAGI score, indicating strong practical potential for routine biological screening and laboratory-based analytical applications.

These advanced tools collectively provide a comprehensive and accurate assessment of the method's greenness and applicability. The combined outcomes of ComplexGAPI, ComplexMoGAPI, AGREEPrep, and BAGI indicate that the PET-CQD-based detection method has promising sustainability and practical applicability within the tested analytical scope.

Although the present strategy offers a waste-to-value route by converting discarded PET into functional nanomaterials, the use of concentrated H_2SO_4 as the reaction medium requires appropriate safety precautions and hazardous waste handling. Therefore, the sustainability advantage of the present approach is primarily associated with PET valorization and the generation of useful sensing materials from plastic waste.

3.9 Comparison with reported Fe^{3+} sensors and practical applicability

To highlight the advantages of the current material over previously reported Fe^{3+}-responsive systems, the sensing performance of PET-CQDs was compared with selected carbon-dot-based and nanomaterial-based probes (Table 1). Table 1 serves to position the present PET-derived CQDs within the broader field of Fe^{3+} fluorescence sensing. Accordingly, the comparison includes sensors prepared from different precursor sources to demonstrate that waste PET can act as a viable feedstock for the development of functional Fe^{3+}-responsive nanomaterials. The proposed sensing system for detecting Fe^{3+} exhibits several notable benefits:

(1) Sustainable and environmentally friendly approach: the sensor is fabricated using recycled waste materials, promoting plastic waste reduction through a single-step preparation method. Its greenness assessment supports its sustainability-oriented profile.

(2) High sensitivity and accuracy: the sensor serves as a fluorescent probe with a suitable limit of detection (LOD), ensuring reliable Fe^{3+} detection in the tested samples. Its ability to detect low levels enhances its applicability in preliminary biological screening and analytical applications.

(3) Ease of use and fast detection: the smartphone-based system enables visual detection without the need for complex instrumentation, providing a rapid and straightforward analytical approach.

(4) Enhanced stability and reliability: the sensor maintains strong performance across various environmental conditions, ensuring consistent and reproducible detection results.

(5) Cost-effectiveness: a brief cost consideration was included only as an illustrative estimate to compare the accessibility of the proposed PET-CQD platform with a conventional instrument-based method such as atomic absorption spectroscopy (AAS). This comparison is not intended as a full techno-economic or life-cycle cost analysis. Based on assumed

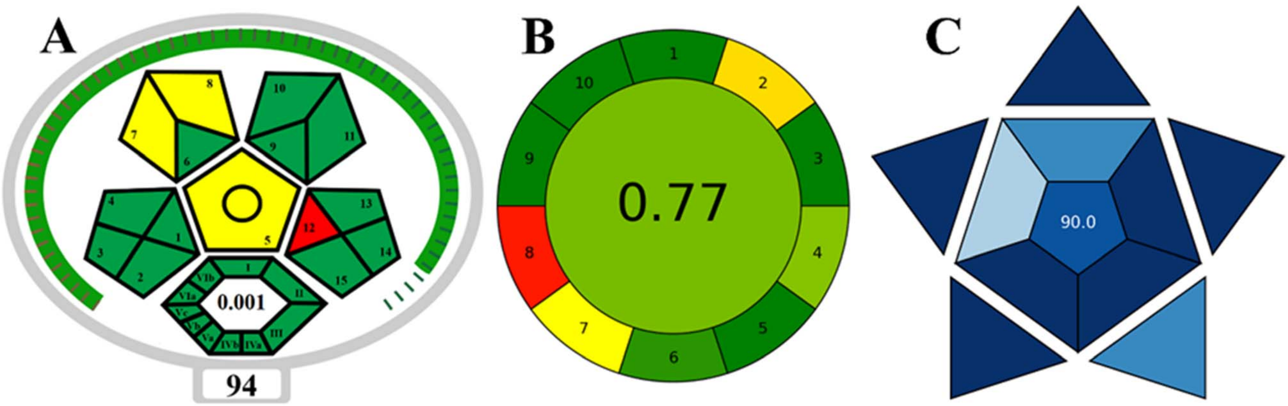


Fig. 6 Greenness evaluation for the proposed method using (A) ComplexMoGAPI, (B) AGREEPrep, and (C) BAGI scales.

Table 1 Comparison of the analytical performance of the current and previously reported methods for Fe^{3+} detection

Target	Sensor	LOD	Linearity	Ref.
Fe^{3+}	NCQDs/sSiO$_2$@mSiO$_2$/8-HQ	0.96 μM	—	113
Fe^{3+}	CQDs	12 μM	0–160 μM	114
Fe^{3+}	AgNPs	25 μM	—	115
Fe^{3+}	o-HBA AuNPs	9.19 μM	—	116
Fe^{3+}	$Ca_{0.05}Eu_{0.01}Y_{1.94}O_3$ nanophosphor	~ 63.2 nM	10–90 μM	117
Fe^{3+}	PPET-CDs	2.14 μM	5–50 μM	65
Fe^{3+}	PET-CQDs, spectrofluorometric detection	3.66 μM	10–100 μM	Current work
Fe^{3+}	PET-CQDs, smartphone-assisted detection	4.4 μM	10–70 μM	Current work

approximate values, including an AAS instrument cost of ~$30 000, annual maintenance of ~$2000, 10 000 lifetime tests, and ~$15 per test for consumables and labor, the estimated cost is about $20.3 per test. In comparison, a CQD-based system using low-cost disposable consumables and a ~$1000 portable fluorimeter amortized over the same number of tests gives an estimated cost of about $2.8 per test. These values are approximate and may vary depending on instrument model, location, maintenance requirements, operator cost, sample throughput, and laboratory infrastructure. Therefore, the comparison should be interpreted as a practical accessibility estimate, suggesting that the PET-CQD platform may offer a lower-cost option for preliminary Fe^{3+} screening, while AAS and related methods remain necessary for validated confirmatory analysis.

These features make the proposed sensing system a practical proof-of-concept platform for Fe^{3+} detection, offering a low-cost and accessible approach for analytical screening.

To clarify the relationship between this work and our previous study, the present manuscript is framed as a controlled extension of PET-derived carbon-dot sensing rather than a duplicate study. While both studies use PET waste as the precursor, the present work employs H_2SO_4-mediated synthesis and produces PET-CQDs with different physicochemical and analytical characteristics, including a smaller average particle size, altered optical behavior, and a different Fe^{3+} response range. In addition, the current work extends the sensing platform through smartphone-assisted image analysis and formal greenness evaluation, which provide practical and sustainability-focused dimensions beyond the earlier PET-CD system.

Although the detection limit obtained in this work for Fe^{3+} is higher than those reported for some state-of-the-art nanomaterial-based sensors,[113] the present platform was designed to emphasize green synthesis, low cost, and smartphone-based portability rather than record-low sensitivity. From this perspective, the practical value of the method lies in its accessibility and potential use as a rapid screening tool. For Fe^{3+}, the LOD is below the lower end of commonly reported serum concentration ranges,[118] suggesting possible utility for preliminary assessment in physiologically relevant concentration windows. Nevertheless, further validation in real biological matrices is necessary before any clinical application can be established. Therefore, the present study should be viewed as a proof-of-concept for a sustainable and portable Fe^{3+}-responsive sensing platform, with future optimization required to further improve sensitivity and expand practical applicability.

4 Conclusions

This study demonstrated the successful conversion of waste PET plastic into fluorescent carbon quantum dots (PET-CQDs) for the detection of Fe^{3+} ions. The synthesized PET-CQDs exhibited a pronounced fluorescence quenching response toward Fe^{3+}, achieving detection and quantification limits suitable for preliminary biological screening applications. The sensing platform was evaluated using both conventional fluorometric measurements and smartphone-assisted visual analysis, confirming its potential as a simple, portable, and cost-effective probe for Fe^{3+} detection. The quenching behavior was investigated using steady-state fluorescence and UV-Vis spectroscopy, and the results support a plausible interaction between Fe^{3+} and PET-CQDs. The probe was examined in deproteinized serum as a proof-of-concept matrix study, where fluorescence quenching was observed under the tested conditions. Nevertheless, comprehensive validation involving a larger number of authentic serum samples, comparison with certified reference methods, and more extensive statistical evaluation is required before routine clinical or diagnostic implementation could be considered. Greenness and practical applicability assessments revealed that the developed method possessed a favorable sustainability profile, primarily owing to the valorization of PET waste, operational simplicity, and the use of smartphone-assisted detection. However, the requirement for concentrated H_2SO_4 necessitated appropriate safety measures and responsible waste management practices. Future studies should therefore focus on enhancing both the analytical performance and the overall environmental sustainability of the proposed sensing platform.

Ethical statement

Serum samples were obtained as anonymized residual specimens from routine clinical laboratory testing. No personal identifiable information or clinical data were collected or used in this study. The samples were handled anonymously and used only for method-evaluation purposes following local institutional procedures for residual clinical specimens. No formal ethics approval number or approval letter is available for this work.

Author contributions

L. J. M.: conceptualization, data curation, methodology, writing – original draft. K. M. O.: supervision, validation, writing – review and editing.

Conflicts of interest

The authors declare no competing interests.

Data availability

The data, all raw data and results, that support the findings of this study are available from the corresponding author upon request.

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